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Archival Report

Identifying Neurophysiological Markers of Intermittent Theta Burst Stimulation in Treatment-Resistant Depression Using Transcranial Magnetic Stimulation–Electroencephalography

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ABSTRACT

BACKGROUND: Intermittent theta burst stimulation (iTBS) targeting the left dorsolateral prefrontal cortex is effective for treatment-resistant depression, but the effects of iTBS on neurophysiological markers remain unclear. Here, we indexed transcranial magnetic stimulation–electroencephalography (TMS-EEG) markers, specifically, the N45 and N100 amplitudes, at baseline and post-iTBS, comparing separated and contiguous iTBS schedules. TMS-EEG markers were also compared between iTBS responders and nonresponders.

METHODS: TMS-EEG was analyzed from a triple-blind 1:1 randomized trial for treatment-resistant depression, comparing a separated (54-minute interval) and contiguous (0-minute interval) schedule of 2 × 600-pulse iTBS for 30 treatments. Participants underwent TMS-EEG over the left dorsolateral prefrontal cortex at baseline and posttreatment. One hundred fourteen participants had usable TMS-EEG at baseline, and 98 at posttreatment. TMS-evoked potential components (N45, N100) were examined via global mean field analysis.

RESULTS: The N100 amplitude decreased from baseline to posttreatment, regardless of the treatment group ($F_{1,106}$ = 5.20, p = .02). There were no changes in N45 amplitude in either treatment group. In responders, the N100 amplitude decreased after iTBS ($F_{1,102}$ = 11.30, p = .001, $p_{corrected}$ = .0004). Responders showed higher posttreatment N45 amplitude than nonresponders ($F_{1,94}$ = 4.11, p = .045, $p_{corrected}$ = .016). Higher baseline N100 amplitude predicted lower post-iTBS depression scores ($F_{4,106}$ = 6.28, p = .00014).

CONCLUSIONS: These results provide further evidence for an association between the neurophysiological effects of iTBS and treatment efficacy in treatment-resistant depression. Future studies are needed to test the predictive potential for clinical applications of TMS-EEG markers.

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Treatment-resistant depression (TRD) is defined as a major depressive episode (MDE) without improvement after adequate courses of pharmacotherapy or psychotherapy (1,2), affecting ~30% of individuals with major depressive disorder (MDD) (3). Repetitive transcranial magnetic stimulation (rTMS) is an effective intervention for TRD (4–6). A potential rTMS mechanism of action is the alteration of cortical excitation or inhibition. Intermittent theta burst stimulation (iTBS) is a Food and Drug Administration–approved rTMS paradigm, appearing to have more pronounced neurophysiological modulation than conventional rTMS (7). These pronounced effects appear to be a result of iTBS pattern, whereby the theta rhythm is coupled with the gamma frequency, a pattern usually involved in learning and memory (8–10). When applied to the left dorsolateral prefrontal cortex (L-DLPFC), iTBS has approximately 50% response and 32% remission rates in TRD (11); however, its precise neurophysiological effects remain unclear.

The combination of investigational TMS and electroencephalography (TMS-EEG) indexes cortical activity from regions without otherwise direct output, including the DLPFC (12–15). Single-pulse TMS-EEG produces TMS-evoked potentials (TEPs) with reproducible peaks (i.e., N45, P60, and N100), thought to be linked to cortical inhibition and excitation (16,17). The N45 and N100 components appear linked to inhibitory $GABA_A$ (gamma-aminobutyric acid A) receptor ($GABA_AR$) and $GABA_BR$ subtypes, respectively (18–21). The P60 component is associated with glutamate, a main excitatory neurotransmitter (8,22,23). The area under the curve (AUC) of the TEP waveform measured by global mean field amplitude (GMFA) (GMFA-AUC) reflects cortical excitability

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(24,25). Notably, MDD pathophysiology is linked to GABAergic deficits (26), associated with cortical inhibition, through multiple lines of research: postmortem (27), magnetic resonance spectroscopy (28–30), multiple TMS cortical inhibition paradigms (31,32), and TMS-EEG (33). Thus, there is much value in clarifying therapeutic rTMS' impact on TMS-EEG markers linked to GABAergic receptor activity (i.e., N45, N100).

The neurophysiological effects of rTMS have only recently been explored in MDD with TMS-EEG. Reductions in P30, N45, and N100 amplitudes followed high frequency left and bilateral DLPFC rTMS (15,34). While P30's significance remains unclear (34), N100 decreased as MDD symptoms improved, suggesting possible therapeutic mechanisms (15). In contrast to conventional rTMS, iTBS administers gamma frequency (∼50 Hz) multipulse bursts at theta rhythm (∼5 Hz), altering neurophysiological mechanisms involved in processes including long-term potentiation, learning, and memory (8–10). iTBS appears to have more pronounced neurophysiological effects than other rTMS paradigms, which require elucidation.

Here, we present results from the largest TMS-EEG dataset to date collected from individuals undergoing iTBS for TRD. TMS-EEG was collected during a triple-blind randomized clinical trial comparing separated (two 600-pulse sessions, 54 minutes apart) and contiguous (two consecutive 600-pulse sessions, 0 minutes apart) iTBS schedules (35) for 30 treatment days. TMS-EEG was recorded at baseline and following 30 treatments of iTBS. We focused on globally extracted TMS-EEG markers to determine cumulative iTBS effects over the cortex, as opposed to immediate effects over the stimulation site. The effect of treatment schedule on N45 and N100 amplitudes was examined to test our hypothesis that the separated iTBS schedule would result in greater changes in N45 and N100 amplitudes than the contiguous schedule (36,37). TEP components in responders and nonresponders were compared. Responders were hypothesized to have greater change in N45 and N100 amplitudes after iTBS. Lastly, associations were explored between TEP components and improvement in depressive symptoms.

METHODS AND MATERIALS

Recruitment and Trial Design

As part of a triple-blind randomized controlled trial conducted at the Centre for Addiction and Mental Health, University Health Network, and the University of British Columbia, participants from Centre for Addiction and Mental Health and the University of British Columbia underwent TMS-EEG at baseline and post-iTBS (clinicaltrials.gov identifier NCT02729792) (35). Inclusion criteria were ages 18–59 years, meeting MDD criteria from the Mini-International Neuropsychiatric Interview, baseline Hamilton Rating Scale for Depression (HRSD-17) score ≥18, no psychotropic medication changes in the 4 weeks prior to screening, and not having met clinical response for ≥1 adequate trial of antidepressant pharmacotherapy or not tolerating ≥2 separate trials of antidepressants. The protocol was approved in accordance with the Declaration of Helsinki; all participants provided written informed consent.

The parent trial consisted of daily iTBS for 6 weeks (30 treatments, 5 days/week) applied over the L-DLPFC. Participants were randomized to either two iTBS sessions of 600 pulses 0 minutes apart (contiguous) or 54 minutes apart (separated). Sham sessions ensured blinding in each schedule (Figure 1). Sessions were performed at 120% resting motor threshold (RMT), with 3 pulse bursts at 50 Hz, repeated at 5 Hz, duty cycle of 2 seconds on and 8 seconds off for 600 total pulses, lasting 189 seconds. For both TMS-EEG and iTBS, the L-DLPFC was specified by reverse coregistration onto each participant's magnetic resonance image using Montreal Neurological Institute coordinates (x −38, y +44, z +26) (38). For further treatment details, including a detailed CONSORT (Consolidated Standards of Reporting Trials) diagram specifying participant recruitment, exclusion, and completion, please see Blumberger *et al.* (35).

Investigational TMS Procedures

One hundred monophasic TMS pulses were administered to the L-DLPFC (Brodmann area 9/46) using a 7 cm figure-eight coil and 2 Magstim 200 stimulators (Magstim Company Ltd.) connected via a BiStim module.

RMT was assessed prior to the first iTBS treatment and at both TMS-EEG sessions (i.e., baseline and post-iTBS). RMT was determined over the left motor cortex, defined as the minimum intensity eliciting motor evoked potentials >50 μV in ≥5 of 10 trials (39), determined both before and after EEG cap positioning. Each session was performed at 120% RMT. TMS-EEG data were collected within 48 hours before starting iTBS and 7 days after completion. All participants underwent identical TMS-EEG measures, with planned pre-post within-subject and within-group analyses.

EEG Recording and Analysis

EEG was recorded using a 64-channel cap, with electrodes on the outer side of each eye and above and below the left eye to monitor eye movement. Electrodes were referenced to a vertex electrode posterior to Cz. EEG signals were recorded direct current at 20 kHz sampling rate and with a low pass filter of 300 Hz to avoid amplifier saturation and minimize TMS-related artifacts (40).

EEG processing was performed by an investigator blinded to treatment arm, using EEGLAB (41) and custom MATLAB scripts (R2013a; The MathWorks, Inc.). Data were epoched around the TMS pulse (−1000 to 1000 ms), baseline corrected (−1000 to −110 ms), and pulse signatures (−2 to 20 ms) were removed and linearly interpolated. EEG data were downsampled to 1 kHz and visually inspected for excess noise (i.e., bad electrodes and muscle movements) (42). A round of independent component analysis removed TMS tails and muscle artifacts (17), followed by bandpass filtering (1–50 Hz). A second round of independent component analysis removed artifacts, including eyeblinks and eye movements. The data were rereferenced to the average electrode. Details regarding dataset inclusion and exclusion are in Figure 1.

GMFA Analysis. A whole-brain approach was taken to quantify neurophysiological response using GMFA, using equation 1, adapted from Lehmann and Skrandies (43). The

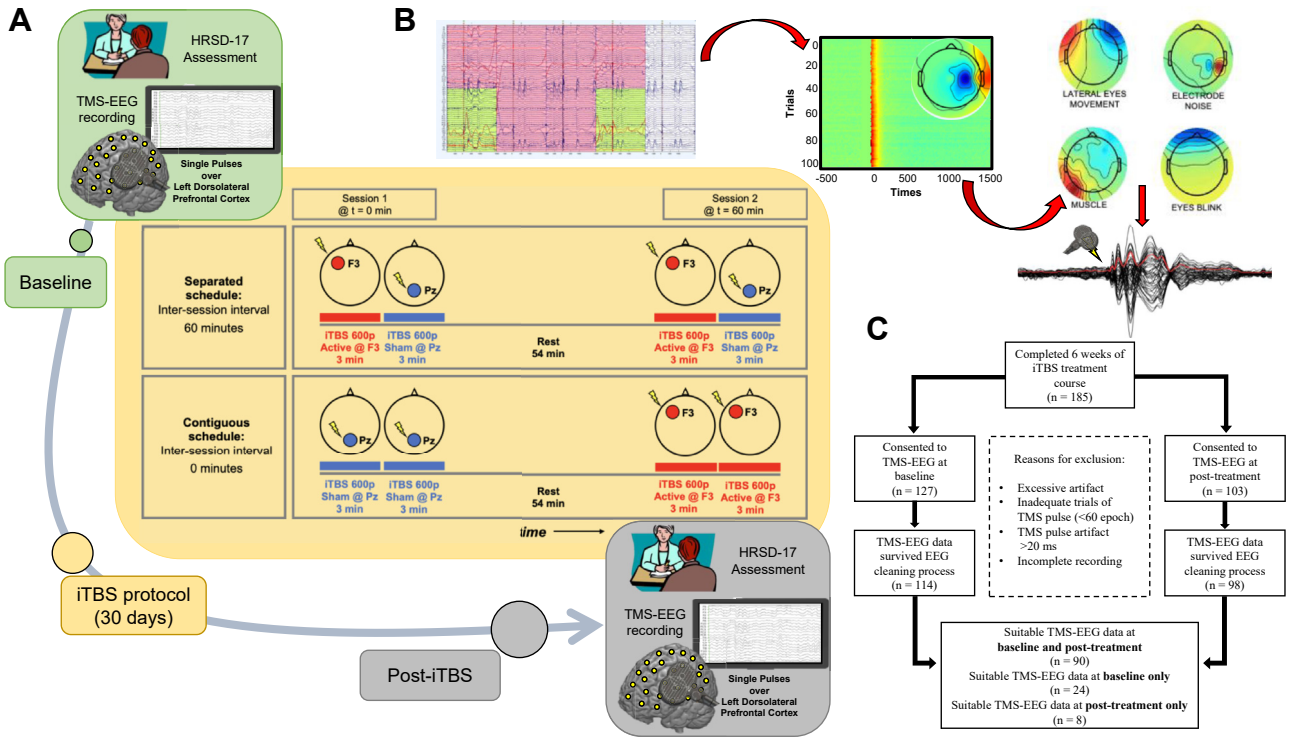


Figure 1. Overview of the study design. **(A)** Parent study design. Participants were randomized to 30 treatments of either 2 × 600 pulse intermittent theta burst stimulation (iTBS) sessions 0 minutes apart (contiguous) or 54 minutes apart (separated), with sham sessions ensuring blinding. Single-pulse transcranial magnetic stimulation–electroencephalography (TMS-EEG) and Hamilton Rating Scale for Depression-17 items (HRSD-17) were performed at baseline and post-iTBS (day 30). **(B)** Overview of TMS-EEG processing steps. First was preprocessing, next was two rounds of independent component analysis to reject eye blinks, lateral eye movements, muscle artifacts, and electrode noise, followed by a visual quality check of each TMS-evoked potential for any leftover artifacts. **(C)** Flow diagram of included TMS-EEG participants.

area under the TEP (GMFA-AUC) from all viable electrodes quantified combined brain response to TMS (24,25,43) (equation 1).

$$GMFA(t) = \sqrt{\sum_{i}^{K} \frac{(Vi(t) - Vmean(t))^2}{K}} \quad (1)$$

GMFA component peak amplitudes and latencies were calculated and visually inspected prior to analysis. Peak amplitudes were calculated for respective time ranges post-TMS pulse: N45 (35–55 ms), N100 (80–120 ms). GMFA-AUC was calculated by determining the sum of the TEP amplitude from 55 ms to 275 ms.

Clinical Variables of Interest

The primary clinical outcome of interest was change in HRSD-17 from baseline to treatment end. Independent raters were blinded to the treatment group. Categorical outcome variables included response (≥50% reduction in HRSD-17) and remission (posttreatment and 1 week follow-up HRSD-17 <8).

Statistical Analysis

Statistical analyses were conducted with R statistical software. Odds ratios (ORs) assessed differences between treatment groups for proportions of females, left-handed subjects, responders, remitters, and medications. Two-sample *t* tests assessed differences between treatment groups in mean age, years of education, MDD age of onset, episode duration, and baseline and posttreatment HRSD-17 scores.

Neurophysiological Outcome Analysis. Correlational analyses were performed between TEP components (N45 and N100) and numeric demographic variables, including age, RMT, sleep duration, ethnicity, education, age of onset, and episode length, as well as N100 TEP component latency. Mann-Whitney *U* tests compared TEP components between categorical variables, including treatment group, sex, and benzodiazepine, caffeine, alcohol, or nicotine use. Mann-Whitney *U* tests examined differences in ΔTEP amplitudes (post-iTBS minus baseline) between treatment and response groups for paired subjects with baseline and posttreatment recordings. Paired sample Wilcoxon tests identified amplitude changes after iTBS within treatment and response groups. Linear mixed models examined changes in N45 and N100 amplitudes between treatment (time × treatment) and responder groups (time × response). Amplitudes were log-transformed to meet model assumptions. Post hoc pairwise comparisons were performed to identify model interactions with Bonferroni corrections to account for multiple comparisons.

Clinical Outcome Analysis. Logistic regression analyses were performed between GMFA component amplitudes (i.e., N45 and N100 amplitudes and GMFA-AUC) and response status. To determine which variables to include in the regression model, Spearman's or Pearson's correlation analyses (as appropriate) were performed between GMFA components and outcome variables (i.e., HRSD-17, sex, and age). Significant correlations underwent subsequent multiple regression analyses using HRSD-17 as the continuous outcome measure and significant correlations as predictor variables. An exploratory bootstrapping analysis was performed between baseline N100 amplitude and posttreatment HRSD-17 score (with 10,000 bootstrapped samples) to determine the 95% CI of correlation coefficients.

Exploratory Analysis of Variability. Given the variability in iTBS response, as well as TMS-EEG results, an exploratory analysis regarding direction of change was undertaken. This included participants with both baseline and posttreatment recordings examining variability in N100 amplitude change after iTBS, split by responders (n = 34) and nonresponders (n = 56). N100 amplitude change was classified as "positive" (>20% increase), "negative" (>20% decrease), or "no change," according to the literature (37). Chi-squared analysis examined differences between proportions of responders and nonresponders in positive, negative, and no change categories.

RESULTS

Demographics and Clinical Variables

One hundred twenty-seven participants consented to baseline TMS-EEG, and data from 114 were retained after EEG cleaning. One hundred three consented to posttreatment TMS-EEG, and data from 98 were retained after EEG cleaning. Ninety participants had both baseline and posttreatment recordings after EEG cleaning. The separated group included 54 baseline and 46 posttreatment, with 42 paired, recordings. The contiguous group included 60 baseline and 52 posttreatment, with 48 paired, recordings.

The TMS-EEG sample was demographically and clinically representative of the parent study (35) (Table 1). There were no significant differences between treatment groups in age, sex distribution, baseline HRSD-17 score (t_{115} = 1.81, p = .07), posttreatment HRSD-17 score (t_{116} = −1.44, p = .15), or proportion of remitters (OR = 1.73, p = .14). Although not statistically significant, the proportion of responders appeared higher in the separated schedule group (OR = 2.09, p = .06). There were higher proportions of benzodiazepine and stimulant use in the separated group (benzodiazepines: OR = 2.63, p = .03; stimulants: OR = 5.69 p = .03). Correlational analyses showed that posttreatment HRSD-17 score was related to the number of iTBS treatment days (ρ = −0.31, p = .001), length of MDE (ρ = 0.22, p = .02), and benzodiazepine use (U = 873.50, p = .04).

Table 1. Patient Demographics and Clinical Characteristics

Characteristics	Separated Schedule, n = 58	Contiguous Schedule, n = 64	t (p Value)	OR (p Value)
Demographics				
Age, Years	41.26 ± 10.28	40.92 ± 12.14	0.17 (.87)	–
Female	68.97% (40)	60.94% (39)	–	1.42 (.35)
Left-Handed	15.52% (9)	6.25% (4)	–	2.76 (.11)
Education, Years	17.19 ± 2.34	17.95 ± 2.28	−0.09 (.93)	–
Age of Onset, Years	20.00 ± 9.25	23.27 ± 10.33	−1.84 (.07)	–
Episode Duration, Months	44.78 ± 57.35	40.78 ± 52.88	0.40 (.69)	–
Clinical				
No. of Failed Adequate Trials	2.24 ± 1.24	2.35 ± 1.42	−0.44 (.66)	–
Baseline HRSD-17	23.57 ± 4.23	22.25 ± 3.77	1.81 (.07)	–
Posttreatment HRSD-17	13.46 ± 7.15	15.30 ± 6.89	−1.44 (.15)	–
Responders	43.86% (25)[a]	26.98% (17)[b]	–	2.09 (.06)
Remitters	22.81% (13)[a]	17.19% (11)[b]	–	1.73 (.14)
Medications				
Benzodiazepine	32.76% (19)	15.63% (10)	–	2.63 (.03)[c]
Antidepressant	39.66% (23)	34.38% (22)	–	1.25 (.55)
Antidepressant Combination	18.97% (11)	23.44% (15)	–	0.76 (.55)
Antipsychotic Augmentation	22.41% (13)	23.44% (15)	–	0.94 (.89)
Antipsychotic Alone	0% (0)	0% (0)	–	
Stimulant	15.52% (9)	3.13% (2)	–	5.69 (.03)[c]
Lithium Augmentation	3.45% (2)	3.13% (2)	–	1.11 (.92)

Continuous variables are denoted as mean ± SD. Categorical variables are denoted as the % (n) of participants in each group. Responders, ≥50% reduction on HRSD-17; remitters, HRSD-17 < 8 at 1 week posttreatment.

HRSD-17, Hamilton Rating Scale for Depression-17 item; OR, odds ratio.

[a] n = 57.

[b] n = 63.

[c] p < .05.

Table 2. Neurophysiological Variables in the Separated vs. Contiguous Schedules

Variables	Separated Schedule, *n* = 58	Contiguous Schedule, *n* = 64	Mann-Whitney *U* Test (*p* Value)
Motor Threshold, %			
Baseline	52.16% ± 8.61%	52.55% ± 9.47%	1524.50 (.81)
Post	53.37% ± 7.70%	52.77% ± 11.29%	1259.00 (.66)
Mean GMFA Components			
N45 Amplitude, μV			
Baseline	3.20 ± 1.64	3.13 ± 1.39	1586.00 (.85)
Post	3.45 ± 1.59	3.17 ± 1.24	1313.00 (.41)
ΔN45	0.15 ± 1.87	−0.04 ± 1.74	1097.00 (.47)
N45 Latency, ms			
Baseline	47.52 ± 5.18	47.90 ± 5.40	1544.00 (.67)
Post	46.59 ± 5.10	48.31 ± 5.22	968.00 (.10)
ΔN45	−0.88 ± 7.43	0.31 ± 8.07	925.00 (.50)
N100 Amplitude, μV			
Baseline	4.47 ± 1.65	3.93 ± 1.48	1889.00 (.13)
Post	3.83 ± 1.28	3.67 ± 1.35	1299.00 (.47)
ΔN100	−0.73 ± 1.66	−0.20 ± 1.91	854.00 (.21)
N100 Latency, ms			
Baseline	106.61 ± 13.72	108.73 ± 11.93	1471.00 (.40)
Post	107.61 ± 12.65	105.94 ± 12.97	1257.50 (.66)
ΔN100	1.26 ± 16.67	−2.19 ± 17.12	1064.00 (.63)
GMFA-AUC, μV			
Baseline	707.09 ± 275.95	696.52 ± 201.68	1572.00 (.79)
Post	671.42 ± 201.86	675.28 ± 208.85	1204.00 (.95)
ΔGMFA-AUC	−47.46 ± 311.00	−3.53 ± 260.99	944.00 (.59)

Variables are denoted as mean ± SD. Delta calculated as post minus baseline component in paired subjects (interrupted schedule: *n* = 42; uninterrupted schedule: *n* = 48).

AUC, area under the curve; GMFA, global mean field amplitude.

Table 3. Neurophysiological Variables in the Responder vs. Nonresponder Groups

Variables	Responders, *n* = 42	Nonresponders, *n* = 78	Mann-Whitney *U* Test (*p* Value)
Motor Threshold, %			
Baseline	53.42% ± 8.66%	52.17% ± 9.00%	1309.00 (.57)
Post	54.17% ± 9.38%	52.40% ± 9.94%	958.50 (.25)
Mean GMFA Components			
N45 Amplitude, μV			
Baseline	3.42 ± 1.87	3.05 ± 1.26	1334.00 (.52)
Post	3.89 ± 1.52	2.96 ± 1.24	696.00 (.002)[a]
ΔN45	0.40 ± 2.04	−0.17 ± 1.61	765.00 (.12)
N100 Amplitude, μV			
Baseline	4.66 ± 1.74	3.96 ± 1.44	1133.00 (.06)
Post	3.44 ± 1.23	3.92 ± 1.34	1375.00 (.06)
ΔN100	−1.21 ± 1.70	0.01 ± 1.73	1339.00 (.001)[a]
GMFA-AUC, μV			
Baseline	773.15 ± 288.37	664.12 ± 201.02	1110.00 (.045)[a]
Post	709.05 ± 264.23	652.81 ± 159.04	1085.00 (.82)
ΔGMFA-AUC	−77.01 ± 370.44	8.13 ± 214.41	1130.00 (.14)

Variables are denoted as mean ± SD. Delta calculated as post minus baseline component in paired subjects (responders: *n* = 34; nonresponders: *n* = 56).

AUC, area under the curve; GMFA, global mean field amplitude.

[a]$p < .05$.

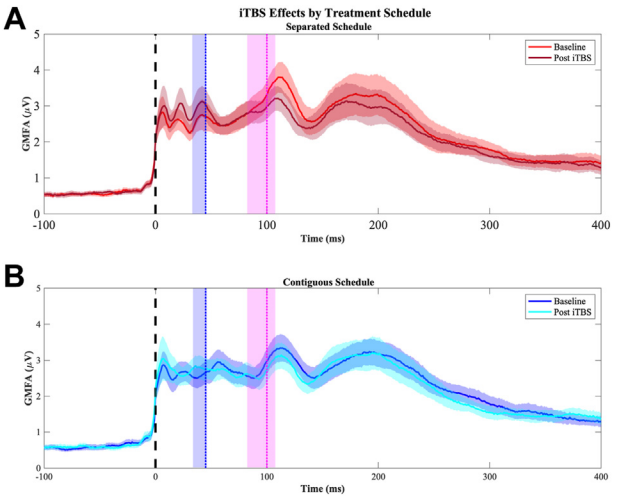


Figure 2. Global mean field amplitude (GMFA) at baseline and after intermittent theta burst stimulation (iTBS) in the **(A)** separated iTBS schedule and **(B)** contiguous iTBS schedule. The black dotted line denotes time 0 (transcranial magnetic stimulation pulse). The blue dotted line denotes 45 ms post–transcranial magnetic stimulation pulse. The blue shaded region denotes N45 time range of 35–55 ms. The magenta dotted line denotes 100 ms post–transcranial magnetic stimulation pulse. The magenta shaded region denotes N100 time range of 80–120 ms. There was no difference in N45 or N100 amplitude from baseline to post-iTBS, or at either time point, between the separated and contiguous iTBS schedules.

Neurophysiological Variables

There was no significant difference in RMT between treatment groups at baseline (U = 1524.50, p = .81) and posttreatment (U = 1259.00, p = .66) (Table 2) or response groups at baseline (U = 1309.00, p = .57) and posttreatment (U = 958.50, p = .25) (Table 3). There was no change in RMT from baseline to posttreatment in either separated (V = 286.50, p = .23) or contiguous (V = 357.00, p = .35) schedules or responder (V = 204.50, p = .40) or nonresponder groups (V = 470.00, p = .22). A scatterplot of baseline versus post-iTBS RMT showed no difference between responders and nonresponders (Figure S1). There was no difference between baseline N45 amplitude (U = 1586.00, p = .85) or latency (U = 1544.00, p = .67), N100 amplitude (U = 1889.00, p = .13) or latency (U = 1471.00, p = .40), or GMFA-AUC (U = 1572.00, p = .79) and no difference in ΔN45 amplitude (U = 1097.00, p = .47) or latency (U = 925.00, p = .50), ΔN100 amplitude (U = 854.00, p = .21) or latency (U = 1064.00, p = .63), or ΔGMFA-AUC (U = 944.00, p = .59) between treatment groups (Table 2).

GMFA Analysis

N45 Component Amplitude. There was no change in N45 amplitude from baseline to posttreatment overall (V = 2024.00, p = .93), in separated (V = 406.00, p = .57) or contiguous (V = 628.00, p = .69) groups. Accounting for RMT in a linear mixed model (r = 0.28, p = .005), there was no difference in N45 amplitude at baseline or post-iTBS between treatment groups ($F_{1,97}$ = 0.04, p = .83, $\eta_p^2 = 4.63 \times 10^{-4}$, d = 0.04) (Figure 2). Mean N45 amplitude posttreatment was higher in responders (U = 696.00, p = .002) (Figure 3). A linear mixed model controlling for RMT showed a significant time × response interaction ($F_{1,94}$ = 4.11, p = .045, η_p^2 = 0.04, d = 0.42), and pairwise comparison with Bonferroni correction revealed higher N45 amplitude posttreatment in responders versus nonresponders ($t_{corrected}$ = −3.049, $p_{corrected}$ = .016) (Figure 4). However, N45 amplitude did not change from baseline to posttreatment in responders ($t_{corrected}$ = −1.99, $p_{corrected}$ = .29) or nonresponders ($t_{corrected}$ = 0.73, $p_{corrected}$ = > .999). There was no difference in ΔN45 between responders and nonresponders (U = 765.00, p = .12). There were no associations between baseline or posttreatment N45 amplitude with clinical variables.

N100 Component Amplitude. N100 amplitude decreased after treatment overall (V = 2556.00, p = .04) and in the separated group (V = 637.00, p = .02) (Figure 2). A linear mixed model demonstrated no significant time × treatment interaction ($F_{1,106}$ = 1.28, p = .26, η_p^2 = 0.01, d = 0.22); in the overall sample, N100 amplitude significantly decreased post-iTBS ($F_{1,106}$ = 5.20, p = .02, η_p^2 = 0.05, d = 0.44) (Figure 5).

N100 amplitude decreased after treatment in responders (V = 495.00, p = .0004) but not in nonresponders (V = 773.00, p = .84) (Figure 3). Interestingly, ΔN100 was different in size and direction between responders and nonresponders (U = 1339.00, p = .01). In responders, ΔN100 became more negative; in nonresponders, it became more positive. A linear mixed model demonstrated a response × time interaction for N100 amplitude ($F_{1,102}$ = 11.30, p = .001, η_p^2 = 0.10, d = 0.67); post hoc testing showed decreased N100 in responders ($t_{corrected}$ = 4.132, $p_{corrected}$ = .0004) but not in nonresponders (Figure 5).

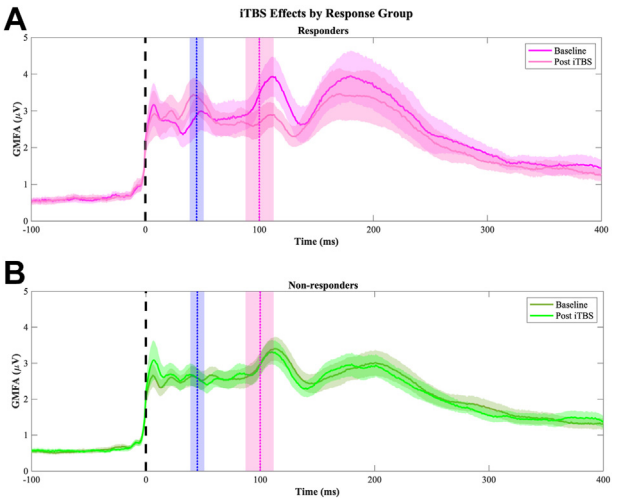


Figure 3. Global mean field amplitude (GMFA) at baseline and after intermittent theta burst stimulation (iTBS) in **(A)** responders and **(B)** nonresponders to iTBS. The black dotted line denotes time 0 (transcranial magnetic stimulation pulse). The blue dotted line denotes 45 ms post–transcranial magnetic stimulation pulse. The blue shaded region denotes N45 time range of 35–55 ms. The magenta dotted line denotes 100 ms post–transcranial magnetic stimulation pulse. The magenta shaded region denotes N100 time range of 80–120 ms. iTBS resulted in decreased N100 amplitude in responders only. Post-iTBS N45 amplitude was larger in responders than in nonresponders.

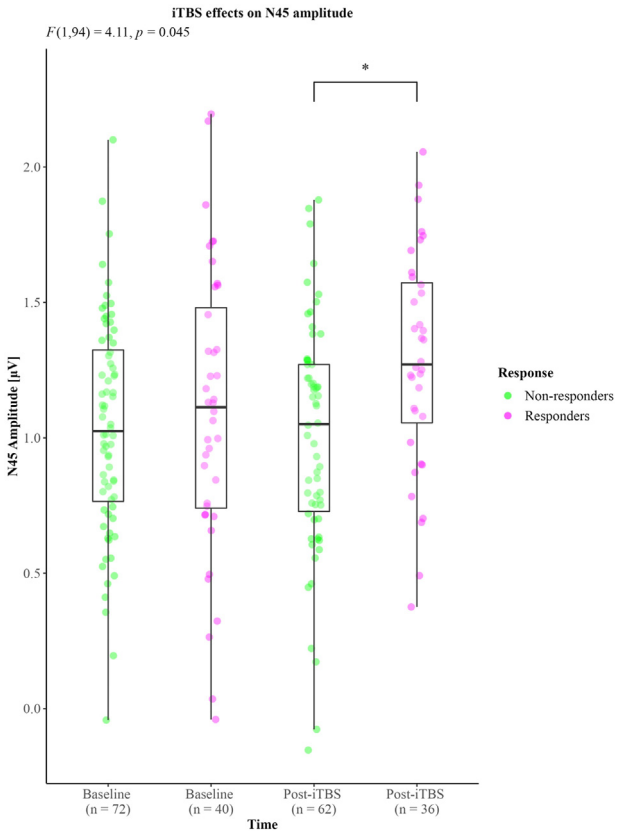


Figure 4. N45 amplitude at baseline and after intermittent theta burst stimulation (iTBS) in responders and nonresponders. $^{*}t_{corrected} = -3.049$, $p_{corrected} = .016$. N45 amplitude was higher in responders at post-iTBS than in nonresponders.

This appeared to be driven by the contiguous group ($F_{1,55}$ = 9.70, p = .003, η_p^2 = 0.15, d = 0.84; $t_{corrected}$ = 3.10, $p_{corrected}$ = .018), with a similar trend in the separated group ($F_{1,44}$ = 2.49, p = .12, η_p^2 = 0.05, d = 0.48).

Higher baseline N100 correlated with lower posttreatment HRSD-17 in the overall sample ($\rho = -0.24$, p = .01; bootstrap 95% CI, −0.4186 to −0.0839) (Figure S2). Higher baseline N100 amplitude predicted lower HRSD-17 scores when accounting for number of treatments, length of MDE, and benzodiazepine use. These variables significantly predicted posttreatment HRSD-17 score ($F_{4,106}$ = 6.28, p = .00014, adjusted R^2 = 0.16). For every 1 μV increase in baseline N100 amplitude, posttreatment HRSD-17 decreased by 0.93 ($t_{106} = -2.13$, p = .035). Decreases in N100 amplitude positively correlated with decreases in HRSD-17 score (r = 0.27, p = .0093). There was no significant association between baseline N100 amplitude and response or remission status.

GMFA-AUC. There was no significant change in GMFA-AUC after iTBS overall (V = 2192.00, p = .56), in separated (V = 521.00, p = .39) or contiguous (V = 588.00, p = > .999) groups. Female participants had higher GMFA-AUC at baseline (U = 2038.00, p = .001). At baseline, participants prescribed benzodiazepines had lower GMFA-AUC (U = 1720.00, p = .04); GMFA-AUC and RMT were positively correlated posttreatment (r = 0.21, p = .04). Accounting for these in a linear mixed model, there was no difference in GMFA-AUC between treatment groups ($F_{1,101}$ = 0.15, p = .70, $\eta_p^2 = 1.45 \times 10^{-3}$, d = 0.07). Responders showed higher baseline GMFA-AUC (U = 1108.00, p = .045). Accounting for sex, benzodiazepine use, and RMT, there were no significant effects of response or time on GMFA-AUC ($F_{1,98}$ = 1.64, p = .20, η_p^2 = 0.02, d = 0.26).

Higher baseline GMFA-AUC correlated with lower posttreatment HRSD-17 ($\rho = -0.25$, p = .007) (Figure S2). Higher baseline GMFA-AUC predicted lower posttreatment HRSD-17 score when accounting for number of treatments, MDE length, and benzodiazepine use ($F_{4,106}$ = 6.05, p = .0002, adjusted R^2 = 0.16). For every 1 μV increase in baseline GMFA-AUC, posttreatment HRSD-17 decreased by 0.006 ($t_{106} = -2.13$, p = .035). There were no associations between baseline GMFA-AUC and response or remission or ΔGMFA-AUC or ΔHRSD-17 in the whole sample.

Exploratory Analysis of Variability. Responders' (n = 34) ΔN100 amplitude was more likely to be classified as negative than that of nonresponders (n = 56) (χ^2 = 4.92, p = .026) (Table S1). Nonresponders' ΔN100 amplitude was more likely positive than that of responders (χ^2 = 7.80, p = .005). There was no difference in the proportion of responders and nonresponders exhibiting no change (χ^2 = 0.14, p = .70).

Benzodiazepine and Stimulant Analysis. There were more participants in the separated schedule prescribed benzodiazepines or stimulants (OR = 2.63, p = .03; OR = 5.69, p = .03, respectively). There was no difference in benzodiazepine or stimulant use between response groups. Excluding individuals prescribed benzodiazepines (separated: baseline = 34 recordings, post-iTBS = 35 recordings; contiguous: baseline = 50 recordings, post-iTBS = 44 recordings) had no effect on TEP marker results. Excluding individuals prescribed stimulants (separated: baseline = 45 recordings, post-iTBS = 39 recordings; contiguous: baseline = 58 recordings, post-iTBS = 50 recordings) had no effect on previously described results. Including benzodiazepine and stimulant use as covariates in linear mixed models did not affect the results.

DISCUSSION

Here, we present evidence of neurophysiological effects of therapeutic iTBS to the DLPFC in TRD, substantiating results in rTMS (15) and MDD (33). iTBS appears to impact inhibitory neurotransmission, specifically inducing an N100 amplitude decrease (Figure 5). This effect was present in the overall sample (Figure 2) but was driven by responders (Figures 3 and 5). There was a consistent decrease in N100 amplitude in responders, whereas nonresponders showed extensive variability (Table S1). Higher baseline N100 amplitude was associated with clinical improvement post-iTBS (Figure S2). N100 amplitude decreased after iTBS concordantly with resolving depressive symptoms, showing synchrony between neurophysiological and clinical responses.

After iTBS, N45 amplitude was higher in responders than in nonresponders (Figure 4), although there was no change in amplitude in either treatment or response group (Figures 2 and

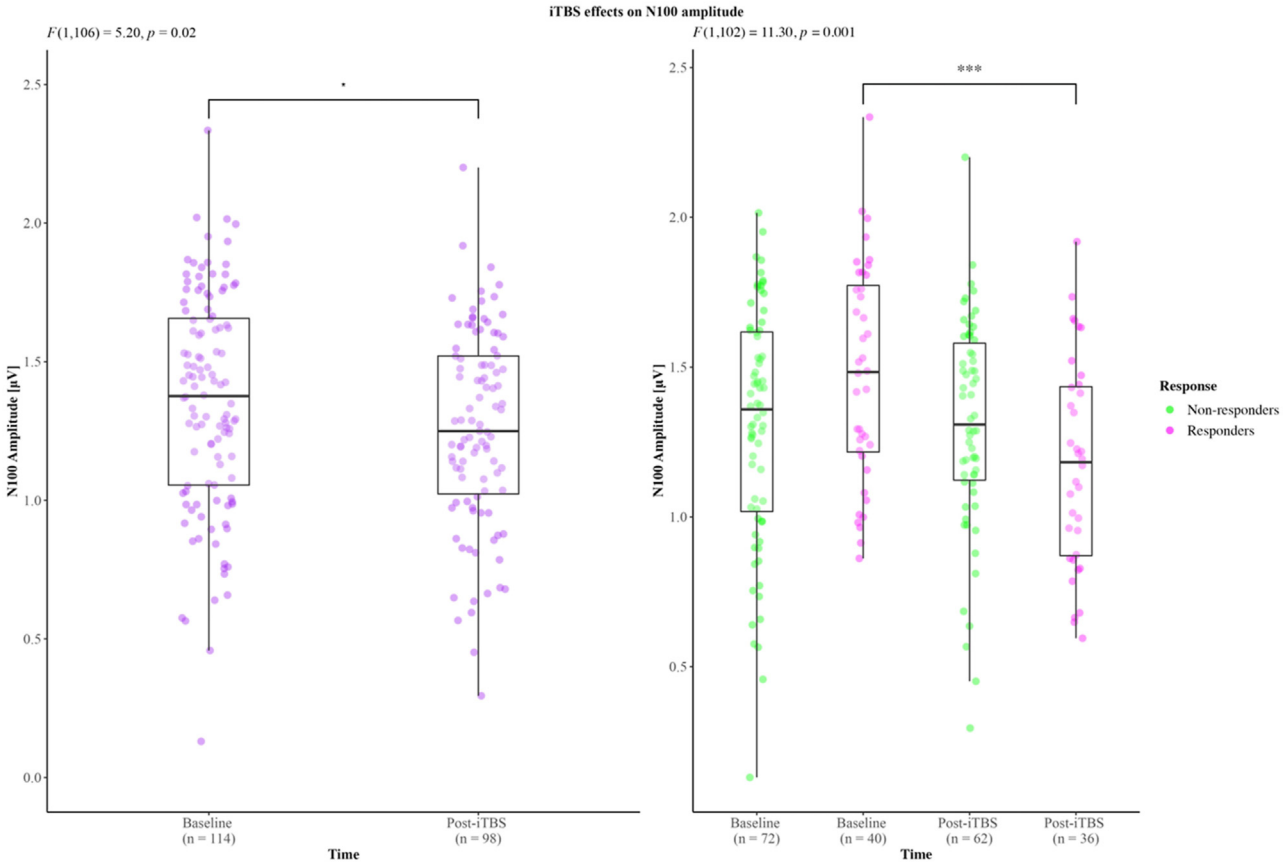


Figure 5. N100 amplitude at baseline and after intermittent theta burst stimulation (iTBS) in the whole sample (left) and responders and nonresponders (right). iTBS resulted in decreased N100 amplitude when collapsed for treatment group. The N100 amplitude decreased from baseline to post-iTBS in the responder group only. $^{*}p < .05$. $^{***}t_{corrected} = 4.132$, $p_{corrected} = .0004$.

3) and no association between N45 amplitude and other clinical variables.

Similarly, there was no change in GMFA-AUC in either treatment or response group. However, participants with higher baseline GMFA-AUC were more likely to have fewer depressive symptoms post-iTBS (Figure S2).

To our knowledge, this study analyzes the largest pre-post TMS-EEG dataset in TRD to date. Other studies had smaller sample sizes and analyzed TMS-EEG following magnetic seizure therapy or electroconvulsive therapy in MDD (n = 38) (44), rTMS in MDD (n = 36) (34), and following single iTBS sessions in stroke survivors (n = 22) (45). This study is, to our knowledge, also the first to examine the effects of therapeutic iTBS over the L-DLPFC using TMS-EEG. Previous mechanistic iTBS studies have largely been limited to the motor cortex, with effects depending on pulses delivered (36,37,46). Here, the total number of daily pulses was consistent across treatment groups (1200 pulses), with a different interval between the two 600-pulse sessions. The interval did not affect TMS-EEG markers, in contrast with motor cortex findings in healthy individuals, where longer intersession intervals increased motor evoked potential size (47,48). One important factor may be the intended stimulation target because the DLPFC may not respond to iTBS in an identical manner to the motor cortex. In fact, in healthy individuals, the N100 did not differ after one or two 600-pulse sessions 15 minutes apart over PFC (48). Inherent differences in cortical regions are evident when examining the effects of therapeutic rTMS on RMT. RMT has previously been shown to increase after rTMS, affecting TEPs and impacting clinical outcomes (49). However, in this trial, there were no changes in RMT following iTBS in the whole sample or response groups (Figure S1). Echoing the parent trial's clinical findings (35), there were minimal TMS-EEG marker differences between treatment schedules, and overall, the N100 amplitude decreased after iTBS. This is in contrast with the neurophysiological differences evident between responder groups, reinforcing the high potential for TMS-EEG indices as markers of iTBS response.

The N100 amplitude decreased only in responders (Figures 3 and 5), aligning with results in more traditional high frequency and sequential bilateral rTMS (15). This decrease may reflect normalization of cortical inhibition, because patients with MDD appear to have higher N100 amplitude than healthy individuals (33). Furthermore, higher baseline N100 amplitude was associated with symptom reduction post-iTBS (Figure S2), echoing the evidence that higher baseline N100 predicts the resolution of suicidal ideation by magnetic

seizure therapy (13). These findings add to the growing body of evidence that the N100 is involved in the pathophysiology of MDD and associated with response to therapeutic brain stimulation.

In contrast, N45 amplitude was higher in responders post-iTBS (Figure 4). The N45 component has been associated with $GABA_AR$ activity (19,22,50) and may be modulated in part by the glutamatergic system (22), while the N100 is linked to $GABA_BR$ activity (19). If N45 amplitude alteration reflects interplay between cortical excitation and inhibition, given susceptibility to both NMDA receptor antagonists and positive allosteric modulators of the $GABA_AR$ (19,22), this may explain the contrasting direction of N45 and N100 in responders. There is clear evidence that cortical imbalance between excitation and inhibition contributes to MDD pathophysiology (26,33,51); therefore, it is possible that the N45 and N100 alterations observed in responders may reflect an iTBS-induced restoration of balance.

Variability in iTBS' impact on the TEP waveform may also be linked to therapeutic response. Responders showed consistent directional change in N100 amplitude, usually negative (Table S1). In contrast, ΔN100 in nonresponders was almost evenly split across positive, negative, and no change classifications (Table S1). Previous studies have placed replicability of expected excitatory or inhibitory effects of TBS paradigms at ≤50% (37,52). Reducing this variability by uncovering its source will help elucidate iTBS' neurophysiological impact and may improve response rates (11,35). While some factors are unmodifiable (52), using advanced targeting methods may account for individualized structure and connectivity, ensuring that the TMS coil is stimulating the intended neuroanatomical structures (53,54).

There were some limitations to this study. First, some patients only consented to TMS-EEG at baseline (n = 127) and others after iTBS (n = 103). By using a mixed model approach, the full dataset was included in the analysis, despite some attrition. However, within-subject differences may have provided important information regarding variability in neurophysiological responses. Second, the separated group included more patients who were prescribed benzodiazepines, which were associated with lower baseline GMFA-AUC and higher posttreatment HRSD-17 score. Benzodiazepines affect motor cortical excitability (55,56), but translation to DLPFC remains unclear. To account for potential benzodiazepine effects, treatment groups were collapsed for response analysis and benzodiazepine use was included as a covariate. Analyses were repeated excluding individuals prescribed benzodiazepines, which did not alter the findings. Third, the difference in the proportion of responders between treatment groups approached significance in the overall sample, thus, response status was included as a covariate. Lastly, no auditory or somatosensory masking was used during TMS-EEG sessions, because part of the parent trial was to test the practicality of TMS-EEG in a clinical setting. Sensory-evoked responses may affect the TEP, especially the N100-P200 complex (57), with larger complexes induced in active versus sham TMS-EEG over the DLPFC (58,59). However, the N100 appears to have a consistent amplitude following suprathreshold stimulation across different anatomical regions, thus reflecting a cortical, not peripheral, source (59). Given that all participants underwent identical TMS-EEG procedures, the decreased N100 amplitude in responders reflects the neurophysiological impact of effective iTBS. Lastly, differences in post-iTBS N45 amplitude between responders and nonresponders, an early TMS-EEG component external to potential auditory confounds (57), provides further robust evidence of iTBS' effects on inhibitory cortical processes linked to GABAergic receptor function, subsequently resulting in antidepressant effects.

Future studies should focus on replicating these findings while taking into consideration the stated limitations. Although the treatment schedule did not alter the neurophysiological effects, different intersession intervals may uncover accumulated plasticity effects (48). Importantly, future research should identify how TMS-EEG markers may help guide TBS. For instance, the current study showed that elevated N100 and GMFA-AUC at baseline predicted improvement in depressive symptoms post-iTBS. The next step could determine a baseline amplitude cutoff to predict the likelihood of response to iTBS. Implementing machine learning algorithms may also increase the accuracy of TMS-EEG biomarkers of response by accounting for other predictors, such as age, refractoriness, severity, and comorbidities (60). There was potential restoration of balance between cortical excitation and inhibition in responders, highlighted by decreased N100 amplitude and higher post-iTBS N45 amplitude. Imbalance of excitation/inhibition has been well characterized in MDD, including TMS-EEG markers (33). The opposing direction of the N45 and N100 following iTBS may indicate a return to normal excitatory and inhibitory neurotransmission, paralleling symptom improvements. Future TRD studies should further investigate such neurophysiological imbalances and alterations with successful intervention. Finally, future TMS-EEG studies should analyze the effects of iTBS on localized and downstream electrodes. Here, the focus was the cumulative iTBS effects after 30 treatment days over the whole cortex using GMFA. However, exploring electrode clusters immediate and distal to the site of stimulation would be an appropriate and informative next step of analysis.

To conclude, this study provides the first evidence of iTBS-induced changes in TMS-EEG markers in TRD in the largest TMS-EEG sample to date. The evidence from this large neurophysiological study shows conclusively that N100 amplitude decreases after iTBS treatment. Interestingly, there was no difference in TEP components whether sessions were delivered contiguously or with a separation. Moreover, the changes seen in N45 and N100 amplitudes are specific to responders, highlighting that iTBS may enact antidepressant effects, in part, through GABAergic alterations. Furthermore, baseline N100 amplitude was associated with iTBS-induced improvements in depressive symptoms and should be examined closely to determine how this measure can be utilized as a reliable predictive neurobiological marker of likely treatment response for patients with TRD in the future.

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